

Medical Insurance Cost Prediction Using Machine Learning

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Abstract—The Medical Insurance Cost Predictor is a machine learning based system developed to estimate the medical insurance cost of an individual from personal, demographic, and health related attributes. Insurance charges can differ considerably according to factors such as age, gender, body mass index, number of children, smoking habits, and geographical region. The proposed approach uses historical insurance data and applies data preprocessing, feature encoding, feature selection, and regression based machine learning techniques to learn the relationship between these attributes and insurance charges. Linear Regression, Decision Tree Regression, and Random Forest Regression are considered for model development and comparison. The dataset is divided into training and testing subsets, and model performance is evaluated using Mean Absolute Error, Root Mean Squared Error, and the R squared score. The trained model can receive the characteristics of a new individual and produce an estimated insurance charge. The system demonstrates how machine learning and regression analysis can be applied to the healthcare insurance domain to provide an approximate cost estimate and to identify factors associated with variations in insurance charges.

Keywords—*medical insurance, machine learning, regression, insurance cost prediction, Linear Regression, Decision Tree, Random Forest, BMI, healthcare analytics*

I. INTRODUCTION

Medical insurance is an important component of financial planning because healthcare expenses can vary widely among individuals. Insurance providers consider several characteristics when determining the premium or expected medical expense associated with an individual. Age, body mass index (BMI), smoking behavior, family dependents, gender, and residential region are among the variables that may be associated with differences in insurance charges. Estimating these charges from historical information is therefore a useful machine learning problem.

Traditional estimation methods generally depend on predefined rules, actuarial assumptions, or manual analysis. Machine learning provides an alternative approach in which patterns present in historical data can be learned automatically. In a regression problem, the model learns a relationship between input variables and a continuous target variable. For medical insurance prediction, the target is the insurance charge, while the input variables represent characteristics of the insured individual.

The objective of this project is to develop a simple and efficient Medical Insurance Cost Predictor using supervised machine learning. The workflow includes collecting the dataset, cleaning and preprocessing the data, converting categorical attributes into a numerical representation, separating input and target variables, splitting the data into training and testing sets, training multiple regression algorithms, and evaluating their predictive performance.

The project also provides an opportunity to examine which characteristics have a noticeable relationship with predicted insurance charges. The resulting system is intended as an educational and analytical prediction tool. Its output represents an estimate based

on the available historical data and should not be interpreted as an actual insurance quotation or a medical recommendation.

II. RELATED WORK

Machine learning based cost prediction has been studied in several domains where a continuous value must be estimated from multiple explanatory variables. Regression algorithms are particularly suitable for such tasks because they model the relationship between independent variables and a numerical target. Simple linear models provide an interpretable baseline, while tree based models can represent nonlinear relationships and interactions among features.

In healthcare analytics, demographic and lifestyle variables are frequently analyzed to understand variation in healthcare expenditure. A medical insurance dataset containing age, sex, BMI, children, smoking status, region, and charges provides a compact example of this type of predictive problem. The project follows a comparative modeling approach so that the performance of a simple linear model can be examined alongside nonlinear tree based methods.

The comparison is important because a model with greater complexity does not automatically guarantee better performance on unseen data. A reliable evaluation therefore requires a separate test set and quantitative error metrics. The proposed work uses MAE, RMSE, and R squared to describe prediction error and goodness of fit.

III. DATASET AND FEATURES

The project uses a tabular medical insurance dataset in which each record represents an individual and contains personal or health related characteristics together with the corresponding insurance charge. The principal input attributes are age, sex, BMI, children, smoker, and region. The target variable is charges, representing the medical insurance cost associated with the record.

Age is a numerical feature representing the individual's age. Sex is a categorical attribute. BMI is a numerical measure related to body composition. Children indicates the number of dependents represented in the record. Smoker identifies smoking status, and region represents the geographical category associated with the individual. The charges attribute is the continuous response variable that the regression models are trained to predict.

These features provide a combination of numerical and categorical information. Because most regression algorithms require numerical inputs, categorical variables must be encoded before model training. Careful preprocessing is therefore an essential stage of the proposed system.

IV. PROPOSED METHODOLOGY

A. Data Preprocessing

The first stage of the proposed methodology is data preprocessing. The dataset is inspected for missing values, duplicated records, inconsistent data types, and invalid entries. Numerical attributes are converted into suitable numerical formats, while

categorical attributes are transformed into machine readable representations. Preprocessing ensures that the training algorithms receive consistent input data.

B. B. Feature Selection

Feature selection identifies the variables used by the predictive model. The proposed system uses the available demographic and lifestyle attributes because they directly describe the characteristics associated with the target insurance charge. The feature matrix contains the selected input variables, while the charges column is separated as the target variable.

C. C. Train Test Split

After preprocessing, the dataset is divided into training and testing subsets. The training subset is used to learn model parameters, while the testing subset is kept separate for evaluating performance on previously unseen records. A fixed random state can be used to make the experimental split reproducible.

D. D. Regression Algorithms

1) Linear Regression: Linear Regression provides a baseline model that estimates the target value as a weighted combination of the input features. It is relatively simple and interpretable and is useful for measuring how well a linear relationship explains the observed insurance charges.

2) Decision Tree Regression: Decision Tree Regression recursively divides the feature space into regions using decision rules. It can represent nonlinear relationships and interactions between variables without requiring the relationship to be explicitly specified as a mathematical linear function.

3) Random Forest Regression: Random Forest Regression combines predictions from multiple decision trees. Each tree is trained using variations of the training data and feature selection, and the final prediction is obtained by aggregating the individual tree predictions. This ensemble approach can model complex relationships while reducing the dependence on a single decision tree.

V. V. SYSTEM IMPLEMENTATION

The system can be implemented using Python and commonly used machine learning libraries. A typical implementation uses pandas for data handling, NumPy for numerical operations, and scikit learn for preprocessing, model training, and evaluation. The workflow begins by loading the insurance dataset and examining its structure. After preprocessing, the input features and target variable are separated and the data is divided into training and testing sets.

Each regression algorithm is trained using the same training data so that the models can be compared under a common experimental setup. Predictions are generated for the test set and compared with the actual charges. The evaluation values are then recorded for model comparison. The selected trained model can subsequently be integrated with a simple user interface in which an individual enters age, sex, BMI, number of children, smoking status, and region to receive an estimated insurance cost.

The overall workflow can be summarized as follows: dataset collection, data inspection, data cleaning, categorical encoding, feature and target separation, train test splitting, model training, prediction, performance evaluation, and deployment of the selected regression model. This sequence makes the system modular and allows additional models or preprocessing techniques to be incorporated later.

VI. VI. PERFORMANCE EVALUATION

The predictive models are evaluated using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R squared (R^2). These metrics provide complementary information about prediction error and the ability of a model to explain variation in the target variable.

Categorical features such as sex, smoker, and region can be represented using one hot encoding. In this approach, each category is converted into a binary indicator so that the regression algorithms can process the information without imposing an artificial numerical order on the categories. Numerical features such as age, BMI, and children are retained as quantitative variables.

A. A. Mean Absolute Error

MAE is the average absolute difference between the actual insurance charges and the predicted charges. A lower MAE indicates that the predictions are, on average, closer to the actual values.

B. B. Root Mean Squared Error

RMSE is the square root of the mean of the squared prediction errors. Because the errors are squared before averaging, RMSE gives greater influence to larger prediction errors. A lower RMSE therefore indicates fewer or smaller large prediction deviations.

C. C. R Squared Score

The R^2 score indicates the proportion of variation in the target variable that is explained by the regression model relative to a baseline model. A value closer to one indicates stronger explanatory performance, although the metric must be interpreted together with error measures and the characteristics of the test data.

D. D. Model Comparison

The three candidate algorithms are compared using the same test set. The comparison should report the MAE, RMSE, and R^2 obtained by each model. The model selection process should consider the complete set of metrics rather than relying on a single measure. Since actual experimental scores depend on the dataset split, preprocessing choices, and model hyperparameters, numerical results should be inserted only after the models have been executed.

VII. VII. RESULTS AND DISCUSSION

The proposed system produces a continuous estimate of medical insurance charges for each test record. The expected output is a numerical cost prediction that can be compared with the corresponding observed charge. The evaluation stage makes it possible to determine how closely each regression algorithm follows the patterns present in the held out test data.

Linear Regression serves as an interpretable baseline and can capture broad relationships between the input features and insurance charges. Decision Tree Regression can capture nonlinear patterns by dividing the feature space into smaller regions. Random Forest Regression extends this approach by combining multiple trees, which can provide more stable predictions for complex relationships. The final comparison should be based on the measured test set metrics rather than assumptions about which algorithm will perform best.

An important analytical aspect of the project is the role of lifestyle and demographic variables. In particular, smoking status can create substantial differences in predicted charges within many insurance datasets, while age and BMI may also contribute to variation. The model learns these relationships from the historical records rather than applying manually specified medical rules. Therefore, feature relationships should be interpreted as associations in the dataset and not as causal medical conclusions.

Because this project is intended as a prediction system, the quality of the output depends on the quality and representativeness of the training data. A model trained on a limited dataset may not generalize equally well to every population, insurer, location, or policy structure. Regular retraining with relevant data and validation on representative samples would be required for a production system.

VIII. VIII. ADVANTAGES AND LIMITATIONS

The proposed system has several practical advantages. It automates the estimation process, accepts multiple demographic and health related inputs, supports comparison of different regression algorithms, and provides measurable evaluation through standard regression metrics. It can also serve as a foundation for a web or desktop application that gives users an immediate approximate estimate.

The system also has limitations. The prediction is dependent on the historical dataset and the variables available in it. Important factors that are not included in the dataset cannot be learned by the

model. Insurance prices may also depend on policy-specific conditions, coverage levels, provider rules, medical history, and market changes that are not represented in a simple public dataset. Consequently, the predicted value should be treated as an estimate rather than a guaranteed insurance price.

IX. IX. FUTURE ENHANCEMENTS

Future work can extend the project by testing additional regression algorithms and tuning their hyperparameters. Cross validation can be used to obtain a more robust estimate of generalization performance. Feature importance and explainability techniques can also be incorporated to help users understand how individual input variables influence predictions.

The system could further be developed as a web application with an interactive form and a prediction dashboard. Additional data sources, larger and more representative datasets, periodic model retraining, and monitoring of prediction errors could improve its practical usefulness. A production version should also include appropriate privacy, security, fairness, and validation procedures because insurance prediction can affect financially significant decisions.

X. X. CONCLUSION

The Medical Insurance Cost Predictor demonstrates the application of machine learning regression techniques to the estimation of medical insurance charges. The proposed workflow combines data preprocessing, categorical feature encoding, train test splitting, regression model development, prediction, and quantitative evaluation. Linear Regression, Decision Tree Regression, and Random Forest Regression provide alternative approaches for learning the relationship between individual characteristics and insurance costs.

The project shows how historical insurance information can be transformed into a predictive system that accepts an individual’s characteristics and produces an estimated charge. MAE, RMSE, and R² provide a structured basis for comparing the models. The system is useful as an educational demonstration of machine learning in healthcare insurance analytics, while real world deployment would require larger representative data, rigorous validation, privacy safeguards, and consideration of insurer specific pricing rules.

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